

Bowenoid Papulosis, Bowen's Disease of the Penis, and Erythroplasia of Queyrat

The terms **Bowenoid papulosis**, **Bowen's disease of the penis**, and **Erythroplasia of Queyrat** remain clinically useful. They describe distinct clinical presentations and reflect a spectrum of penile intraepithelial neoplasia (PeIN) with varying risks of progression to **squamous cell carcinoma (SCC)**. While all may progress to invasive SCC, the risk differs significantly:

- **Bowenoid papulosis**: ~1% risk of progression
- **Bowen's disease of the penis**: ~5% risk
- **Erythroplasia of Queyrat**: 10%–40% risk

These entities represent different phenotypes within the broader category of PeIN.

PeIN of the Balanopreputial Epithelium (Erythroplasia of Queyrat)

- **Typical appearance**: Red, velvety, well-circumscribed lesion on the glans or inner prepuce, typically in uncircumcised men.

PeIN of Keratinized, Hair-Bearing Skin (Bowen's Disease of the Penis)

- **Typical appearance**: Scaly, discrete, erythematous patches or plaques on the shaft skin.

PeIN (Bowenoid Papulosis)

- **Typical appearance:** Clinically resembles genital warts; lesions may be pigmented or erythematous papules or plaques, often grouped.
- **Patient profile:** Typically younger individuals compared to those with Bowen's disease or Erythroplasia of Queyrat.

Diagnosis

- **Biopsy is essential**
- Histology reveals **penile intraepithelial neoplasia**, which can be:
 - **Differentiated type:** Associated with lichen sclerosus
 - **Undifferentiated type:** HPV-associated

Management

Patients with suspected penile cancer or precancer should be managed jointly by **dermatology and urology/andrology** specialists. A combined, sequential approach is often required, tailored to:

- Patient age, comorbidities, and immunosuppression status (e.g., HIV, transplant recipients)
- Anatomical site(s) involved
- Circumcision status
- Underlying pathogenesis (HPV vs. lichen sclerosus)
- Histological subtype (differentiated vs. undifferentiated)

Topical Medical Therapies

- **Imiquimod 5%** (Evidence level: 1,C)
- **Fluorouracil cream 5%** (2,C)
- **Fluorouracil 0.5% / salicylic acid 10% combination** (2,C)

Surgical and Ablative Treatments (aim for tissue conservation)

- **Surgical excision** – Local excision usually sufficient and effective (1,B)
- **Mohs micrographic surgery** (1,B)
- **Cryotherapy** (2,D)
- **Photodynamic therapy** (2,D)
- **Laser ablation** (2,D)
- **Mandatory circumcision** for balanopreputial disease, especially in high-risk patients (e.g., HIV, transplant recipients) (1,D)
- **Glans resurfacing** – Considered if topical treatments fail (2,D)

Adjunctive Measures

- **Polyvalent HPV vaccination** (2,D)
- **Smoking cessation** (2,D)

Follow-up

- **Follow-up is usually mandatory** due to risks of field cancerization and recurrence. Up to one-third of patients may have microinvasive or invasive disease at presentation.
- The optimal duration of follow-up is uncertain.
- **Circumcised patients with Bowenoid papulosis or PeIN confined to the prepuce** may be considered for discharge after successful treatment.
- **Circumcision is strongly recommended** in relevant cases due to the risk of progression and recurrence.